

We conduct interchange intervention experiments on each of the attribute variables, A_{Cont} , $A_{Country}$, and A_{Lang} , to align the language model’s internal representations with the hypothesized causal model. We use two types of counterfactuals: the *attribute counterfactual*, which alters the queried attribute in the base prompt, and the *Wikipedia counterfactual*, which is a freeform sentence from Wikipedia about the entity city. If the MI method successfully isolates the target attribute, then the intervention should cause the language model to output the corresponding attribute value for the entity city in the counterfactual.

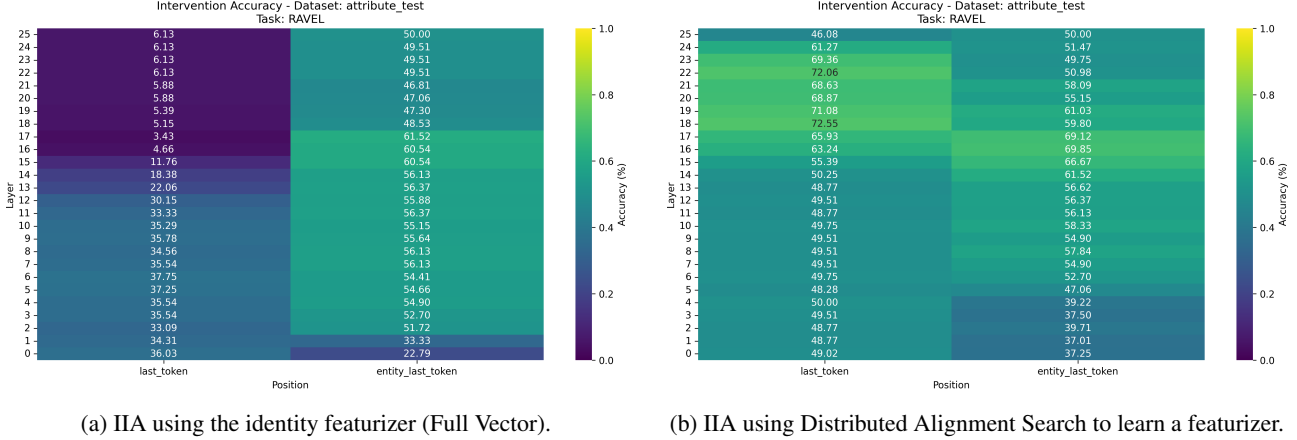


Figure 12. Interchange Intervention Accuracy (IIA) at each layer of Gemma-2 2B, when targeting the $A_{Country}$ variable using the *attribute counterfactual*. The Full Vector baseline fails to isolate the target variable, whereas DAS achieves high accuracy in the mid layers.

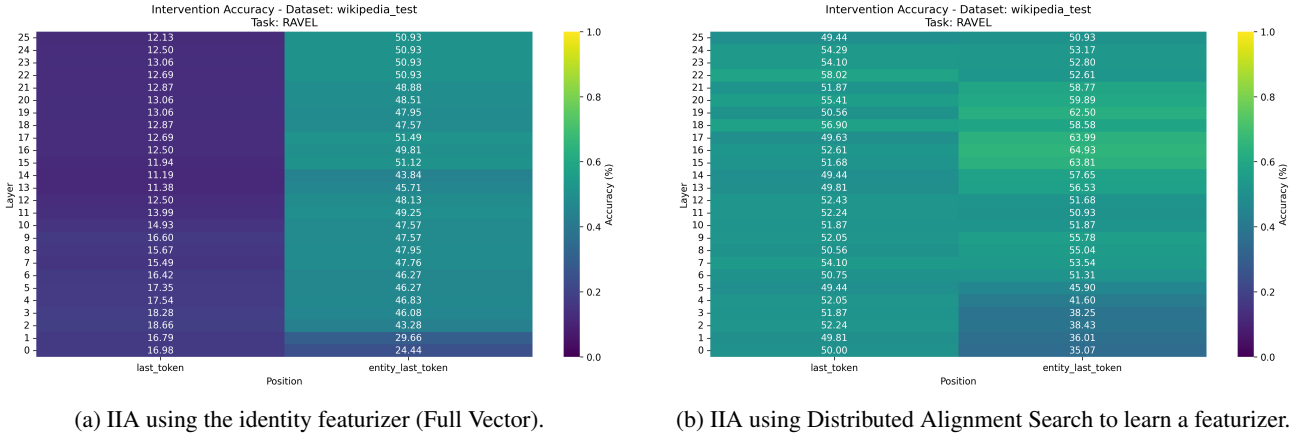


Figure 13. Interchange Intervention Accuracy (IIA) at each layer of Gemma-2 2B when targeting the $A_{Country}$ variable using the *Wikipedia counterfactual*. This counterfactual presents a more challenging setting. The Full Vector baseline fails to disentangle the attributes, while DAS identifies meaningful signal in the mid layers.

Results. We present representative results from the set of intervention experiments. Specifically, we focus on the Gemma-2 2B model and target the $A_{Country}$ variable. Figure 12 shows the Interchange Intervention Accuracy (IIA) at each layer using the *attribute counterfactual*, comparing the baseline (Full Vector) and the best-performing method (DAS). Half of this counterfactual dataset consists of prompts querying the $A_{Country}$ variable, while the other half queries the A_{Cont} or A_{Lang} variables. Successfully isolating the $A_{Country}$ variable requires the method to change only the value of $A_{Country}$ while preserving the values of A_{Cont} and A_{Lang} . Since the Full Vector swaps out all features, it performs poorly in the last token position in the later layers. In contrast, the featurizer learned by DAS achieves high IIA in layers 18 and 19.

We also evaluate performance using the *Wikipedia counterfactual*, which presents a more challenging case: the counterfactual prompt is a freeform sentence that does not query an attribute, and thus would reduce false positives where the counterfactual

answer coincides with the base prompt. In Figure 13, we observe similar but generally lower IIA patterns compared to the *attribute counterfactual*. The Full Vector baseline fails to isolate the variable, with IIA in both token positions consistently around or below 50. For DAS, IIA increases from the early to mid layers, with meaningful signals emerging in the entity’s last token position in the mid layers. It is worth noting that the $A_{Country}$ variable may be more difficult to disentangle than others (Huang et al., 2024a), and all five methods we evaluate exhibit lower IIA on $A_{Country}$ compared to the other two variables.

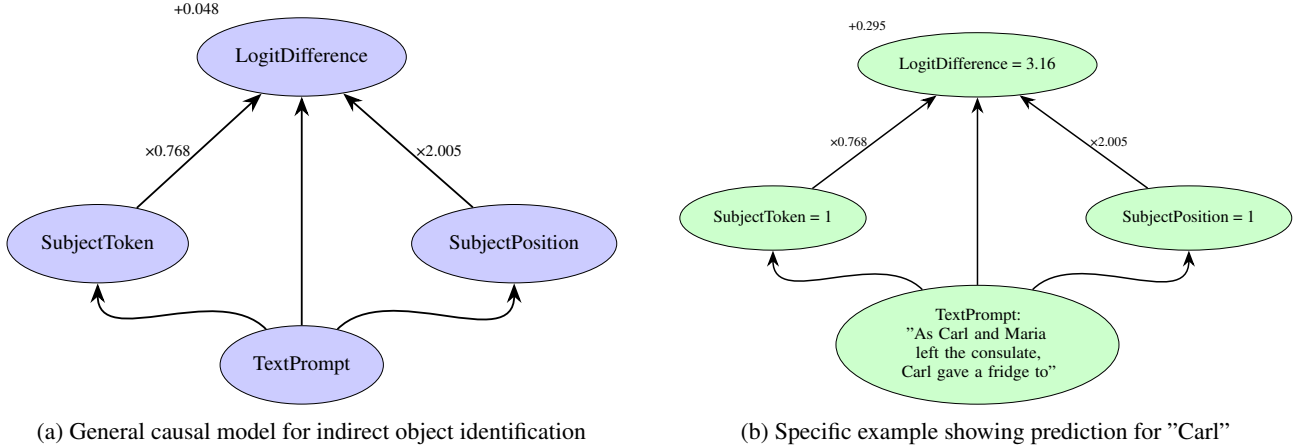


Figure 14. Causal model for indirect object identification. The TextPrompt is processed to extract the subject token S_{Tok} and the subject position S_{Pos} . The output variable mechanism (1) compares the token and position to the input, and determines whether the token and position were inverted or the same and sets PositionSignal and TokenSignal to 1 and -1 accordingly and (2) computes $0.048 + 0.768\text{TokenSignal} + 2.005\text{PositionSignal}$, with position having a stronger influence than token identity.

F.5.4. INDIRECT OBJECT IDENTIFICATION

The indirect object identification (IOI) task is a natural language task which consists of sentences like “When Mary and John went to the store, John gave a drink to”, and evaluates for the model completion, ‘Mary’. The task is linguistically fundamental and has an interpretable algorithm: given two names in a sentence, predict the name that isn’t the subject of the last clause.

A sentence in IOI has two parts: a beginning clause that depends on the rest of the sentence, like “When Mary and John went to the store,” and a main clause, like “John gave a bottle of milk to Mary.” The beginning clause introduces the indirect object (IO), ‘Mary’, and the subject (S), ‘John’. The main clause mentions the subject again, and in every IOI example, the subject gives something to the IO. The goal of the IOI task is to predict the last word of the sentence, which should be the IO. Our high-level model predicts the *logit difference* resulting from subtracting indirect object name logit from the the subject name logit. When this difference is positive, the model is more likely to predict the subject than the indirect object.

(Wang et al., 2023) identify that heads 3 and 9 in layer 7 (7.3 and 7.9) and heads 6 and 10 in layer 8 (8.6 and 8.10) all reduce the likelihood of the model outputting the subject, dubbing these heads “S-inhibition heads”. These heads help the model output the indirect object to solve the task. Specifically, Wang et al. (2023) found that S-Inhibition Heads use two types of signals. The first is the token signal, which carries the token identity of the subject, while the second is the position signal, which carries information about the position of subject, i.e., first or second. These information signals carried by the S-Inhibition heads inform other components in the model to avoid the token and position of the subject.

To identify these two signals, Wang et al. (2023) perform interchange interventions that manipulate each signal separately. The $IO \leftrightarrow S1$ -Flip counterfactual inverts the position of the subject while keeping the token the same. The $IO \leftrightarrow S2$ -Flip inverts the token of the subject while keeping the position the same. The $IO \leftrightarrow S1$ -Flip+ $IO \leftrightarrow S2$ -Flip inverts the position and token of the subject.

After an interchange intervention is performed on all four heads with one of these counterfactuals, Wang et al. (2023) consider the token/position to have value 1 if unchanged and value -1 if inverted.¹³ Using these binary signals as inputs and the logit

¹³They also included a value of 0 for the token signal of a random new token, but we leave out this condition for our experiments.

Table 18. Mean Squared Error (MSE) for aligning S_{Pos} and S_{Tok} to each subset of heads.

Heads	MSE (S_{Pos})	MSE (S_{Tok})
((7, 3), (7, 9), (8, 6))	2.45	6.83
((7, 3), (7, 9), (8, 10))	3.08	6.79
((7, 3), (7, 9))	5.42	3.28
((7, 3), (8, 6), (8, 10))	2.52	8.08
((7, 3), (8, 6))	4.17	4.82
((7, 3), (8, 10))	4.92	3.51
((7, 3))	12.03	4.07
((7, 9), (8, 6), (8, 10))	2.90	10.20
((7, 9), (8, 6))	3.15	5.05
((7, 9), (8, 10))	3.63	4.33
((7, 9))	7.90	2.80
((8, 6), (8, 10))	3.00	5.71
((8, 6))	5.81	4.21
((8, 10))	7.28	2.82

difference between indirect object and subject as outputs, they perform a regression and find that $2.31\text{PositionSignal} + 0.99\text{TokenSignal}$ is the best predictive model of logit difference. On our dataset, we perform the same intervention experiments and fit a linear model, finding that $0.048 + 2.005\text{PositionSignal} + 0.768\text{TokenSignal}$ is the best predictor of logit difference.

Wang et al. (2023) never conducted experiments attempting to disentangle the position and token signals, and this is what we do here.

Dataset. Our dataset consists of train, validation, and test splits. The train, test, and private test sets contain 10000 examples each, as well as 10000 examples for 8 corresponding counterfactuals as shown in Table 6.

Causal Model. We define the causal model for indirect object identification in Figures 14. The variables S_{Tok} and S_{Pos} are derived from the original prompt and without intervention, they will always take on the value of the subject token identity and subject position, respectively. These variables are not equivalent to the binary signals TokenSignal and PositionSignal . Instead, the output O_{LogDiff} variable compares the subject token and subject position to the original input to determine whether the token and position is inverted and run the linear model accordingly.

Without an intervention, the S_{Tok} and S_{Pos} variables will always match the text input, and the logit diff is predicted to be $0.048 + 2.005 + 0.768$. When only S_{Tok} is intervened on, the logit diff is predicted to be $0.048 + 2.005 - 0.768$. When only S_{Pos} is intervened on, the logit diff is predicted to be $0.048 - 2.005 + 0.768$. If both are intervened on, then the logit diff is predicted to be $0.048 - 2.005 - 0.768$.

Full Vector Brute-Force Search We conduct a brute-force search by aligning each of the two causal variables with every possible subset of the four S-inhibition heads. In Table 18 we report the mean-squared error for the alignments. In the main text, we report the alignment of S_{Pos} to heads 7.3, 7.9, and 8.6 and S_{Tok} to the head 8.10.